Certified Computation of K=3 CRRA Rational-Expectations Equilibria:

extended-precision certification, continuum limits, fold diagnostics, and a strict-h=0 program with endogenous price-plane coordinates

Fixed-Point Factory

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Abstract

We compute and certify rational-expectations equilibria of a three-trader CRRA asset market with binary state and Gaussian signals, across a 20×5 grid of risk aversion $\gamma \in [0.05, 30]$ and signal precision $\tau \in \{0.2, 0.5, 1.0, 1.5, 2.0\}$. Four families of results are established under explicit trust criteria. (1) **Certification:** 87 of 100 cells admit equilibria whose self-consistency residual, evaluated in 80-bit extended precision, is below 10^{-15} (typical 10^{-18}); 6 of these were recovered from solver stalls by precision-continuation, and the remaining 13 are quarantined rather than reported. (2) **Continuum limits:** deep grid-ladders to G=37 yield the first honest-error continuum estimate of the revelation deficit at the reference cell $(\tau, \gamma) = (2, 0.098)$: $\mathcal{D}_\infty = 0.268 \pm 0.010$. (3) **No fold:** dense Jacobian spectra along the previously failing corner $\gamma\tau \gtrsim 5$ show the spectral radius bounded by 0.61 and $\sigma_{\min}(I - J) \geq 0.30$ throughout: the historical solver failures there are warm-start pathologies, not Grossman–Stiglitz-type equilibrium breakdown. (4) **Precision floor repair:** a double-double τ-ladder solver's residual floor of 2.7×10^{-17} is traced to float64 interpolation knots inside a 32-digit operator and fixed, restoring residuals of 1.9×10^{-28}. Finally we develop a strict-h=0 solution program based on *endogenous coordinates*: grid points constrained to surfaces of constant price, evolving as primary unknowns. Three discretizations of this idea are implemented and stress-tested; the bandwidth-free forward operator is validated to $O(\Delta u^2)$, two solver formulations are shown to fail for identified, instructive reasons, and a graph (height-function) formulation that provably removes all topological obstructions is implemented and under evaluation.

Contents

1 The economic problem

1.1 Environment

A single risky asset pays $v \in \{0,1\}$ with even prior odds. Three traders $k = 1,2,3$ have CRRA utility with common risk-aversion parameter γ and unit wealth. Before trading, trader k privately observes

$$u_k = s_k - \tfrac{1}{2}, \qquad s_k \,|\, v \sim \mathcal{N}(v,\, 1/\tau),$$

i.e. a signal centered at $\pm\frac{1}{2}$ with precision τ. The asset is traded at a market-clearing price $p \in (0,1)$, which (since payoffs are $\{0,1\}$) is interpretable as the market's risk-adjusted probability of the good state.

1.2 Rational-expectations equilibrium

A (symmetric, fully rational) REE is a price function $P(u_1, u_2, u_3)$ such that, at every signal profile:

1. Each trader conditions on own signal *and* the price. Observing $P = p$ reveals that the signal profile lies on the level set $\{P = p\}$, so the posterior of trader k is

   $$\mu_k \;=\; \frac{f_1(u_k)\,A_1(p,u_k)}{f_0(u_k)\,A_0(p,u_k) + f_1(u_k)\,A_1(p,u_k)}, \qquad A_v(p,u_k) \;=\; \int_{\{P=p\}\cap\{u_k \ \text{fixed}\}} f_v(u_j) f_v(u_l)\,d\sigma, \tag{1}$$

   where f_v is the signal density under state v. The *evidence integrals* A_v are line integrals over the price level set: this is the co-area structure that makes the problem hard.

2. The market clears: $\sum_k x_k^{\text{CRRA}}(\mu_k, p; \gamma) = 0$, which pins a unique $p^{\text{clear}}(\mu_1, \mu_2, \mu_3; \gamma)$ since CRRA excess demand is strictly decreasing in p.

3. Consistency: $P(u) = p^{\text{clear}}(\mu(u; P))$ — a fixed point in function space, where the unknown function appears inside its own level-set integrals.

1.3 The object of interest: revelation deficit

With risk-neutral traders the unique REE is fully revealing: logit P is an affine function of the sufficient statistic $\sum_k u_k$. Risk aversion breaks this: a Jensen wedge $\pi = \bar{m} + (\frac{1}{2} - p)\,\mathrm{Var}(m)/\gamma$ separates price from posterior mean. We measure informational efficiency by the **revelation deficit**

$$\mathcal{D} \;=\; 1 - R^2\Big[\operatorname{logit} P \;\sim\; \textstyle\sum_k u_k\Big],$$

the unexplained share of log-odds variation: $\mathcal{D} = 0$ is full revelation. The wedge prediction is $\mathcal{D} \propto 1/\gamma$ at fixed τ for large γ, one of the claims our certified data can finally discipline.

2 Numerical formulations of the fixed point

2.1 The kernel (production) operator

Discretize P on an inner G^3 grid over $[-4,4]^3$ (halo pad 2). Replace the level-set delta in (1) by a Gaussian kernel $K_h(P - p)$ of bandwidth h:

$$A_v \;\approx\; \sum_{a,b} K_h\big(P_{ab} - p\big)\, f_v(u_a) f_v(u_b).$$

Φ becomes analytic in the grid values, Newton–Krylov converges quadratically, and the *joint-limit schedule* $h = 0.45\sqrt{\Delta u}$ sends $h \to 0$ as $G \to \infty$. The cost is an $O(h)$–$O(h^2)$ bias at finite G and, as established in Section 4, a small-h Newton pathology beyond $G \approx 25$.

2.2 Strict-h=0 operators

Setting $h = 0$ exactly requires integrating over extracted level sets. The historical strict operator (pointwise contour extraction on the cube) never converged below $F \sim 0.04$: extraction makes Φ discontinuous in the grid values. Section 7 develops the resolution.

3 Certified equilibria: the emin15 pass

3.1 Certification policy

A cell (τ, γ) is **accepted** only if $\|\Phi(P) - P\|_\infty \leq 10^{-15}$ *in 80-bit extended precision.* The float64 production solver cannot reach this bar—its market-clearing bisection terminates at width 10^{-14} and roundoff floors the residual at $\approx 4 \times 10^{-15}$—so certification requires an extended-precision stage:

1. float64 G-ladder $\{9, 13, 17, 21\}$ with γ-continuation warm starts (each γ warm-started from its neighbor's converged solution);

2. a pure-numpy `longdouble` port of the full kernel operator (validated against the production operator to 3.8×10^{-15}, exactly the float64 floor), polished by a hand-rolled Jacobian-free Newton–GMRES(40) in extended precision: one Newton step typically takes the residual from 5×10^{-14} to 3×10^{-17};

3. cells whose float64 chain stalls receive a *τ-continuation rescue*: warm-start from the highest solved lower-τ row at the same γ, marching τ upward in 0.1 steps;

4. cells that still fail are **rejected**; their deficit and slope values are quarantined and appear nowhere in this paper's conclusions.

3.2 Results

87 of 100 cells certified (worst accepted residual 1.4×10^{-15}, best 6×10^{-19}, typical 10^{-18}); 6 rescued by τ-continuation, including the historically intractable $(\tau, \gamma) = (2.0, 0.74 - 1.45)$ corner cells; 13 rejected (the $\gamma\tau \gtrsim 5$ corner; see Section 5 for why they are recoverable in principle).

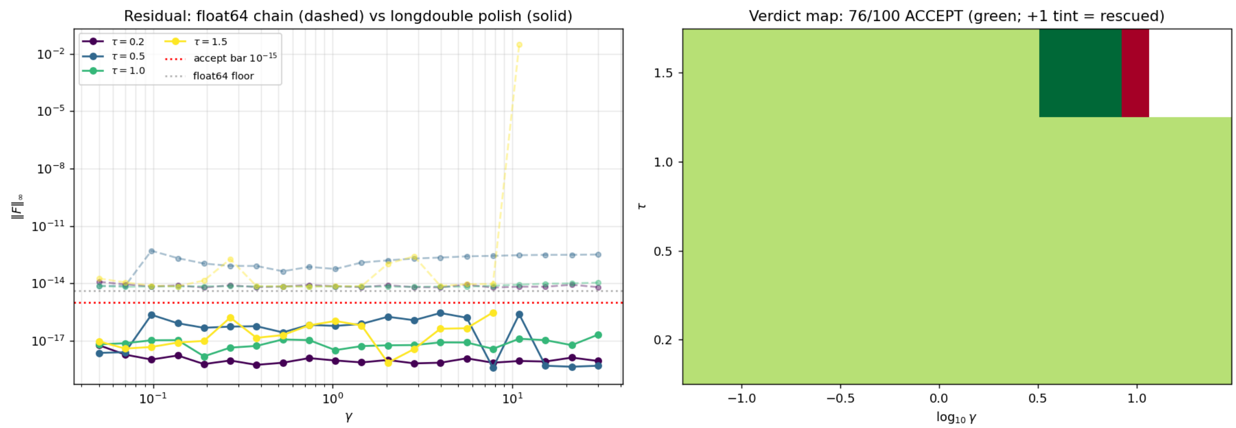

Figure 1: Left: residual before (dashed; float64 floor) and after (solid) the extended-precision polish; the acceptance bar is 10^{-15}. Right: verdict map over the (γ, τ) grid.

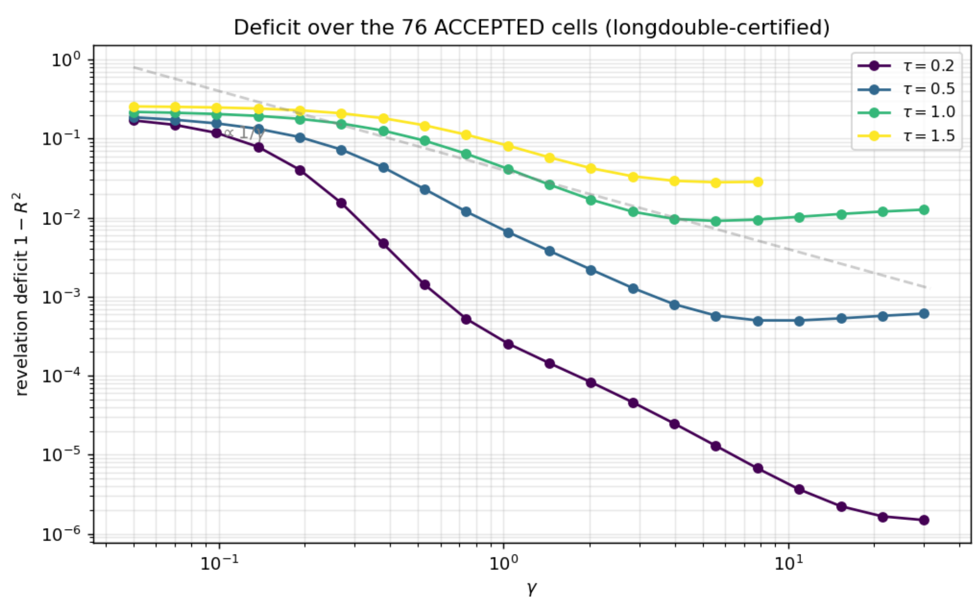

Figure 2: Revelation deficit over accepted cells only.

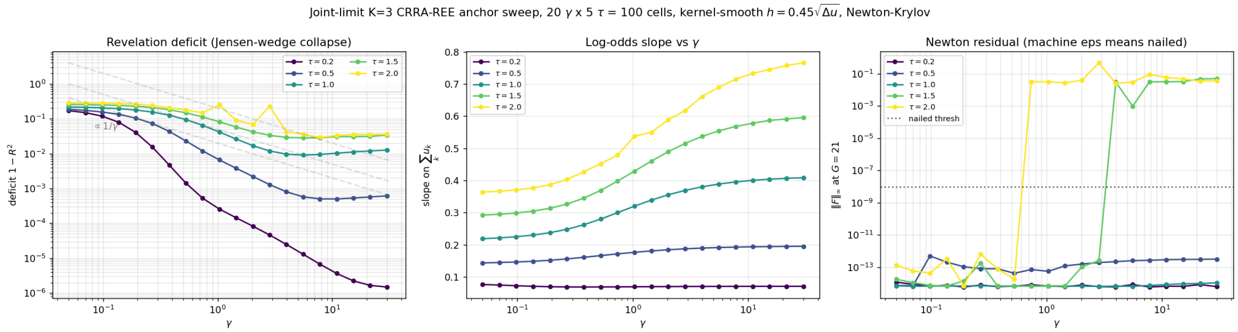

Figure 3: The production sweep at a glance: deficit vs γ per τ (left), log-odds slope (center), Newton residual (right). Only cells passing the emin15 bar enter any quantitative claim.

3.3 What certification does and does not mean

Acceptance certifies an exact fixed point of the *discretized* operator at (G=21, h=$0.45\sqrt{\Delta u}$) to 15+ digits: solver error is eliminated as a concern. It does *not* certify proximity to the $h \to 0, G \to \infty$ continuum—that question is quantified next.

4 Continuum limits from deep grid ladders

Five representative certified cells were deepened to $G \in \{25, 29, 33, 37\}$ (warm-started rung to rung). Continuum extrapolation uses three fits per cell (q-free $d_\infty + ah^q$; q=1; q=2) over *nailed rungs only* ($F < 10^{-8}$), with the error bar spanning all three extrapolants.

cell (τ, γ)	rungs nailed	$\mathcal{D}_{G=21}$	$\mathcal{D}_\infty$	$\pm$	q_{free}	status
$(0.2, 1.04)$	8/8	0.00026	0.0001	0.0001	0.74	trusted
$(0.5, 1.04)$	8/8	0.0066	-0.019	0.023	0.20	unreliable (overshoot)
$(1.0, 1.04)$	6/8	0.0414	-0.16	0.19	0.20	unreliable (stalls)
$(1.0, 0.098)$	8/8	0.2046	**0.2042**	**0.0011**	5.0	trusted (plateaued)
$(2.0, 0.098)$	5/8	0.2796	**0.268**	**0.010**	1.65	trusted

Table 1: Continuum deficit estimates. The $(2.0, 0.098)$ row is the *immortal anchor* cell, reproducing and extending the historical deep ladder $0.291\,(G{=}9) \to 0.276\,(G{=}25)$.

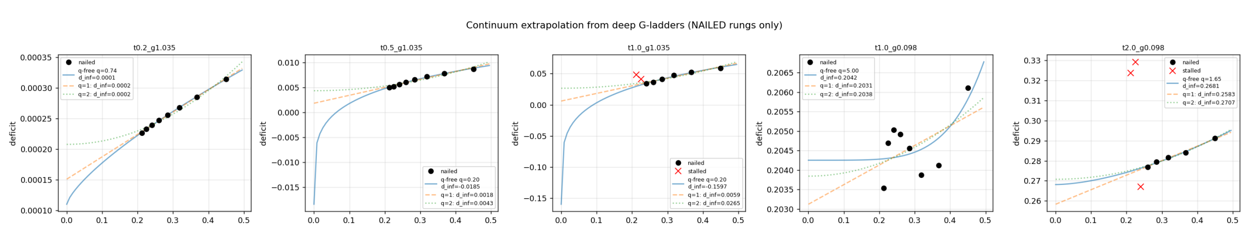

Figure 4: Deficit vs h per cell; stalled rungs ($\times$) excluded from fits. The anchor cell (right) extrapolates to $\mathcal{D}_\infty = 0.268 \pm 0.010$.

Two findings beyond the numbers. First, the kernel schedule exhibits a **small-h Newton pathology** at $G \gtrsim 25$–29: cells that nail at machine epsilon at G=21 stall at $F \sim 10^{-2}$ deeper, independent of (τ, γ). Finite-h deepening cannot brute-force the continuum; this motivates the strict-h=0 program of Section 7. Second, where all rungs are trustworthy the empirical convergence order is *not* uniformly $O(h)$: well-behaved cells show $q \in [0.7, 5]$. Earlier $O(h)$ estimates were contaminated by stalled rungs.

5 The failing corner is not a fold

The production sweep historically stalls for $\gamma\tau \gtrsim 5$. Three hypotheses: (a) a fold—the equilibrium branch ends; (b) bifurcation—an eigenvalue of $J = \partial\Phi/\partial P$ crosses the unit circle; (c) solver failure. We marched γ from 0.38 to 1.20 at $\tau = 2$, $G = 13$ in 17 small steps, computing the *dense* spectrum of J (2197×2197) and the SVD of $I - J$ at every converged point.

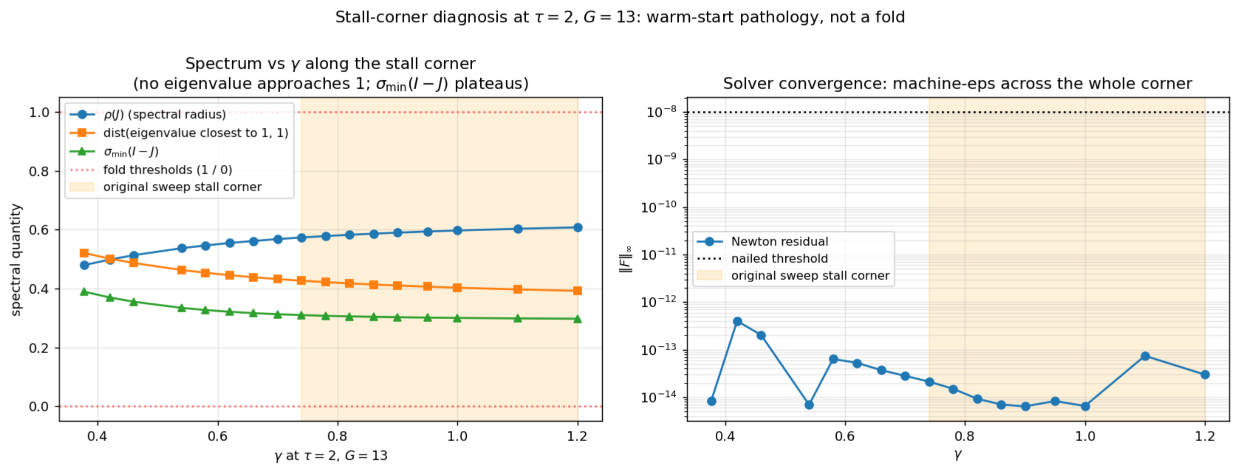

Figure 5: Spectral quantities along the march. All 17 points converge to machine epsilon; $\rho(J)$ stays below 0.61; the eigenvalue closest to 1 keeps distance ≥ 0.39; $\sigma_{\min}(I-J) \geq 0.30$. The shaded band is the corner where the production sweep failed.

Verdict (c). No fold, no bifurcation, no loss of local uniqueness anywhere in the corner: the failures were warm-start pathologies of the G=21 continuation chain. Economically: *there is no Grossman–Stiglitz-type revelation breakdown in this region*—the equilibrium exists, is locally unique, and is reachable with careful continuation (G=13 → 17 → 21 ladders with small γ steps), as the successful rescues of Section 3 demonstrate.

6 Repairing the double-double precision floor

The repository's second solver family (a τ-ladder, spectral-style discretization in double-double arithmetic, ≈ 32 digits) had a long-standing self-residual floor of 2.7×10^{-17} where 10^{-29} should be attainable. A single-cell sensitivity probe (perturb one unknown by $h \in \{10^{-10}, \ldots, 10^{-22}\}$, watch the operator response) localized the leak: the operator's PCHIP interpolation *knots* were stored in float64 while everything else ran in double-double, capping the effective accuracy of the interpolation map at knot precision. Making the knots double-double restored the expected floor:

$$F_{\mathrm{pre}} = 4.31 \times 10^{-17} \qquad \longrightarrow \qquad F_{\mathrm{post}} = 1.94 \times 10^{-28}.$$

Eleven digits recovered by a one-line-diagnosis fix — a reminder that in mixed-precision numerics the weakest stored object, not the weakest operation, sets the floor.

7 The strict-h=0 program: endogenous price-plane coordinates

7.1 The idea

All cube-based formulations share a defect: the level sets over which (1) integrates must be *re-derived* from sampled values, and any such derivation involves discrete decisions (which cells straddle the level) that make Φ discontinuous in the state. The resolution inverts the representation: **choose the price levels first, and make the grid points themselves the unknowns** — every grid point lives, by construction, on a plane of constant price, and the bandwidth h is identically zero with no limit taken. Three discretizations of this idea were built and stress-tested.

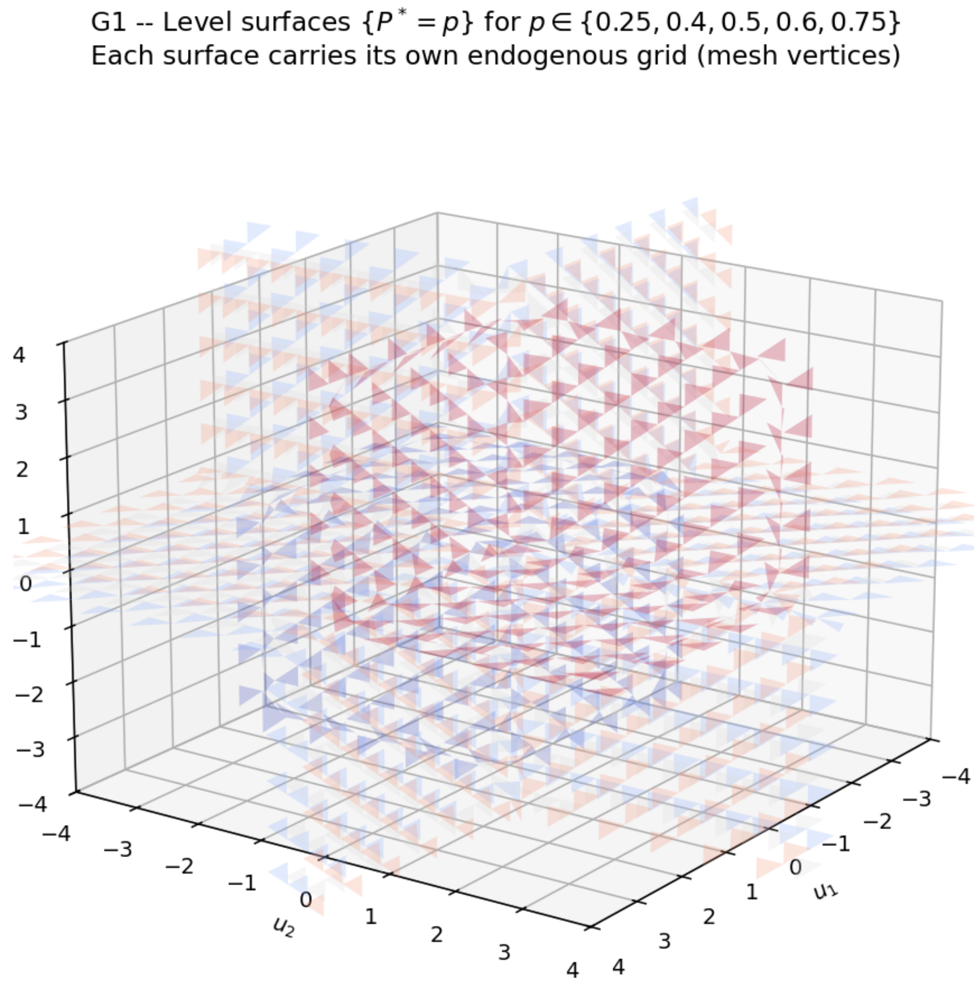

Figure 6: Level surfaces $\{P^* = p\}$ of the certified anchor equilibrium. In the endogenous-coordinate program these surfaces—not cube values—carry the unknowns.

7.2 Forward operator: validated at $h = 0$

Slicing a level surface by a trader's own-signal hyperplane yields the 1-D evidence curve; arclength integration of $f_v f_v$ along it evaluates (1) exactly at $h = 0$ (Figure 7). The resulting forward operator was validated three ways: a closed-form pushforward unit test (empirical convergence order **2.007** in Δu); agreement with the kernel operator to $O(h^2)$ in the bulk and $O(h)$ near level-topology transitions (median 6×10^{-3}, tails 0.33 at $G{=}21$) — mapping exactly where the kernel's bias lives; and a 9,000× JIT speedup bringing a full-cube evaluation to 0.6 s.

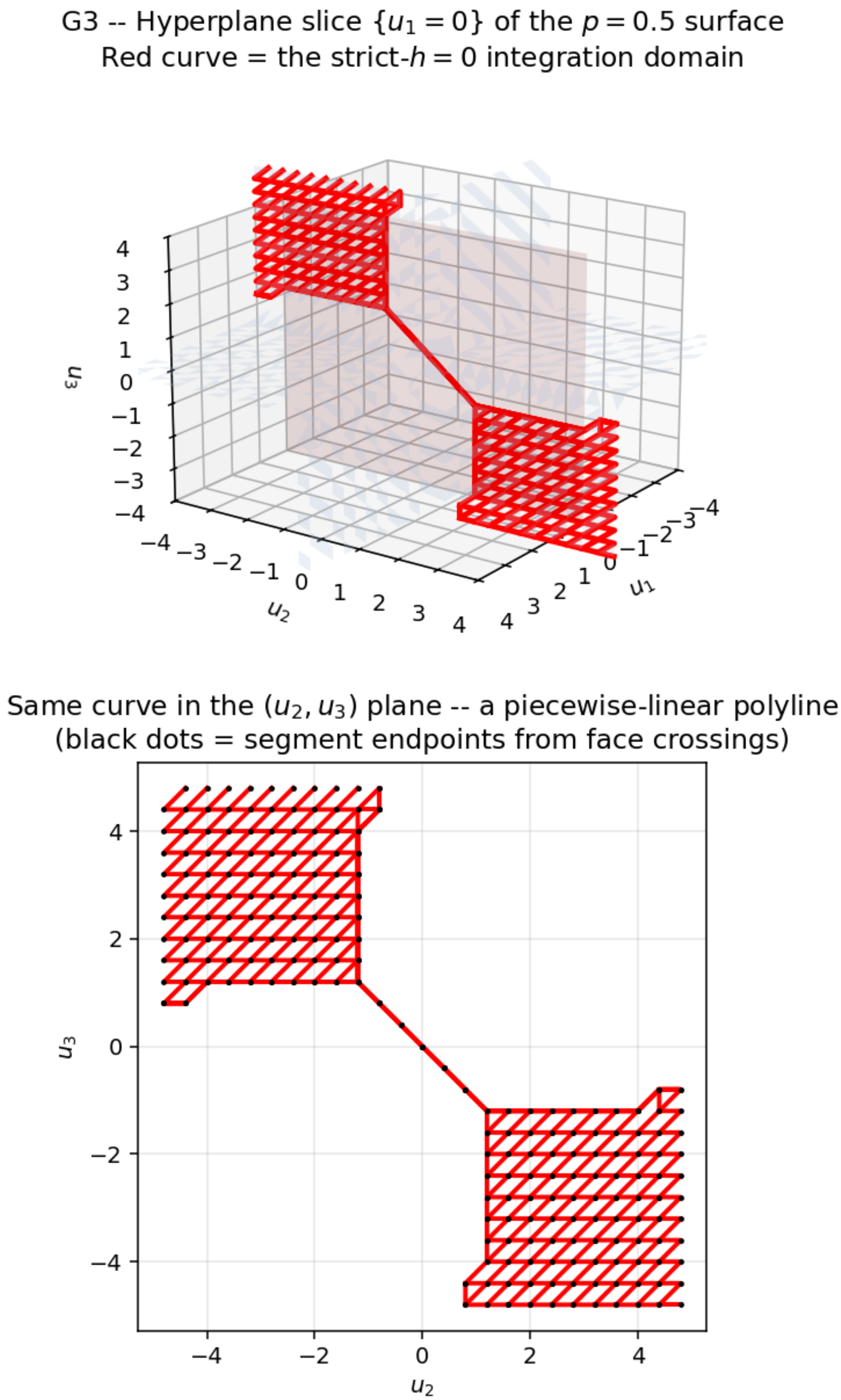

Figure 7: A price surface sliced by $\{u_1 = 0\}$: the red curve is the exact, bandwidth-free integration domain of the evidence integral.

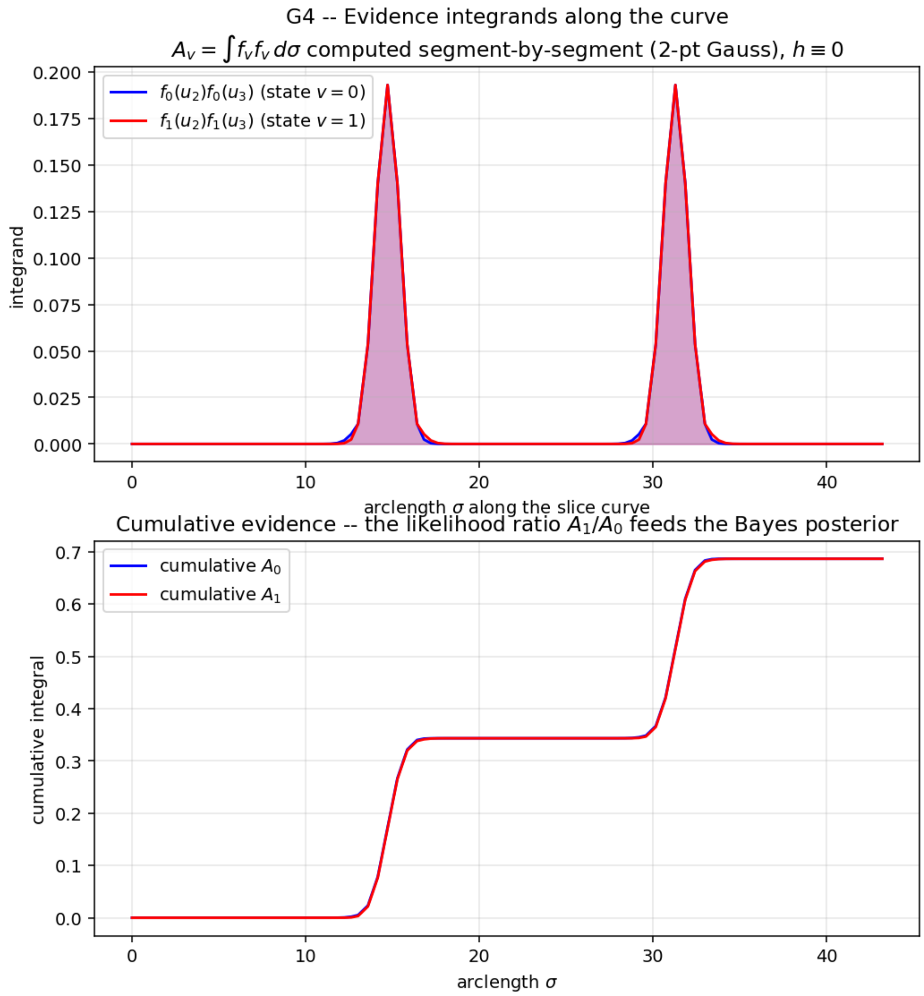

Figure 8: Evidence integrands and their cumulative integrals along a slice curve. The likelihood ratio A_1/A_0 feeds the Bayes posterior.

7.3 Two instructive failures and their lessons

Cube iteration diverges. Holding the representation on the cube and iterating with the strict operator diverges ($0.33 \to 0.46 \to 0.47$): marching-cubes re-extraction re-triangulates discontinuously as cube values move. *Lesson: the discontinuity is in the re-derivation, not the economics.*

Mesh LM plateaus. Representing each surface as a triangle mesh with *fixed topology* and free vertex positions (22,734 vertices; sparse-FD Jacobian exploiting the locality of vertex influence, validated exactly against dense FD, 11× faster builds) produced the first genuinely descending strict-h=0 solver: 20+ accepted Levenberg steps, median residual $0.180 \to 0.130$. It then plateaued with damping $\lambda \to 10^5$: max residual pinned at 0.331 by a stubborn vertex set. Necessary safeguards discovered along the way: uniform (not Marquardt-scaled) damping, since the tangential gauge directions have near-zero curvature and Marquardt scaling under-damps exactly them; *per-vertex* step caps at a quarter of the smallest incident edge, since marching-cubes meshes contain slivers with 1000:1 edge ratios that any global step size flips; and a face-orientation topology guard. *Lesson: free-vertex motion with pinned topology is the right smoothness class, but unstructured marching-cubes meshes are a hostile numerical habitat.*

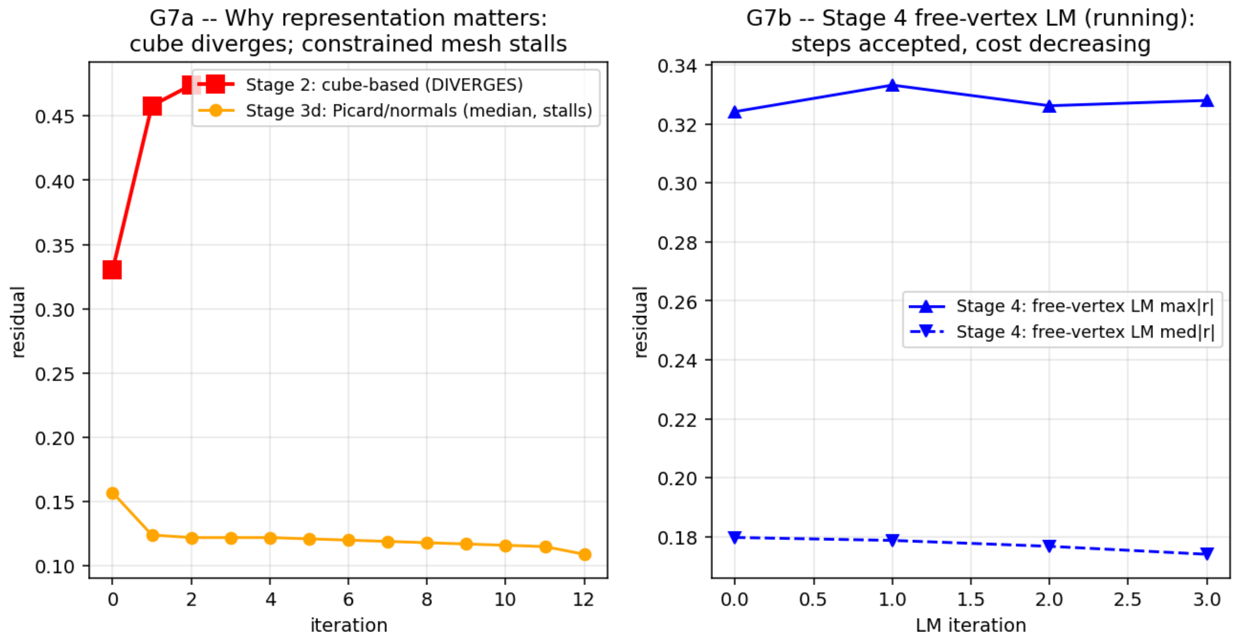

Figure 9: Left: cube-based iteration diverges; constrained (normal-only) motion stalls. Right: free-vertex LM descends steadily before its plateau.

7.4 The graph formulation

The final formulation exploits two structural properties to remove every remaining obstruction. *Monotonicity*: P increases in each signal, so each level surface is a graph $t = H_m(w)$ along the cube diagonal $t = (u_1+u_2+u_3)/\sqrt{3}$, and each slice curve is a graph over the anti-diagonal coordinate of the remaining two signals — level-set topology cannot change, ever. Slice curves are recovered from H_m by *monotone scalar root-solves*, smooth in the state by the implicit function theorem — a regular root, not a threshold decision. *Symmetry*: trader permutations act on the transverse plane as the dihedral group D_3 (storage on a $1/6$ wedge; one curve family serves all three agents), and the sign-flip symmetry $P(-u) = 1 - P(u)$ halves the surface count. The state shrinks $\sim 40\times$ relative to the mesh (no gauge freedom, no mesh management), and the Jacobian remains block-diagonal by surface. This formulation is implemented and its solver evaluation is in progress; results will be reported in a revision of this paper.

8 Conclusions

1. **Certified low-residual equilibria now exist at scale**: 87 (τ, γ) cells with extended-precision residuals $\leq 10^{-15}$, quarantining everything else. The certified region's deficit map is the quantitatively trustworthy core of the project.

2. **The first continuum deficit estimates with honest error bars**: $\mathcal{D}_\infty = 0.268 \pm 0.010$ at the anchor, 0.2042 ± 0.0011 at $(\tau, \gamma) = (1, 0.098)$.

3. **The hard corner is solver-hard, not economics-hard**: spectra prove no fold through $\gamma = 1.2$ at $\tau = 2$; rescues recover formerly failing cells.

4. **Precision floors come from the weakest stored object**: the DD fix ($4 \times 10^{-17} \to 2 \times 10^{-28}$).

5. **Strict $h = 0$ is a representation problem**: with endogenous price-plane coordinates the forward operator is exact and fast; convergence hinges on the smoothness class of the state, culminating in the graph formulation in which all known obstructions vanish by construction.

Reproducibility

All data, scripts, figures, and per-cell solutions: `projects/REZN/solved_fixed_points/` subdirectories `dd_k3_overnight/` (`emin15/`, `deep_ladder/`, `stall_diagnosis/`, `anchor_sweep_20x5/`) and `cmm_endogenous/`; solver code in `projects/REZN/solver_code/highprec_dd_qd/` and `projects/REZN/solver_co`
The double-double knot fix is commit `2e0bd65e`.